Static Replication

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1 Introduction

A general technique to value vanilla options is a dynamic hedging (see, e.g., Black and Scholes, 1973). Theoretically beautiful yet practically unviable scheme of continuous rebalancing that works solely in perfect markets. In practice, practitioner must pick the frequency of rebalancing trading off the transaction costs. Derman, Ergener, and Kani (1995) offer a solution with their *static replication* method. It avoids the rebalancing dilemma by building a fixed portfolio of vanillas that already encodes the tricky features. Intuitively, static replication builds a *fixed-weight portfolio of standard options* (plus cash/forwards) that matches a target option's value across a carefully chosen *boundary* in price-time. By the standard backward induction, if two portfolios have the same boundary values and no intermediate cashflows, they coincide everywhere inside the boundary.

2 Formal Statement

Let $g(S_T)$ be any twice differentiable payoff at maturity T. Pick a reference strike $K_0 > 0$. There is a *static* decomposition using OTM calls/puts, cash, and a forward:[1]

$$
\begin{aligned}
g(S_T) &= g(K_0) + \frac{\partial g(K_0)}{\partial K}(S_T - K_0) \\
&\quad + \int_0^{K_0} \frac{\partial^2 g(u)}{\partial K^2}\,(u - S_T)^+\,\mathrm{d}u \;+\; \int_{K_0}^{\infty} \frac{\partial^2 g(u)}{\partial K^2}\,(S_T - u)^+\,\mathrm{d}u.
\end{aligned}
\tag{1}
$$

Intuitively, each term in (1) corresponds to a basic building block:

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[1]This is the classic static-replication identity for payoffs at a single maturity.

- $g(K_0)$ is a pure constant payoff, which can be replicated by holding a risk-free bond. It anchors the overall level of the portfolio.
- $\frac{\partial g(K_0)}{\partial K}(S_T - K_0)$ is linear in S_T, identical in form to a forward contract with strike K_0. This captures the local slope (first-order sensitivity) of the payoff at the reference strike.
- $\int_0^{K_0} \frac{\partial^2 g(u)}{\partial K^2}(u - S_T)^+ \, du$ is a continuum of puts with strikes $u \le K_0$, weighted by the second derivative of g. These provide downside convexity.
- $\int_{K_0}^{\infty} \frac{\partial^2 g(u)}{\partial K^2}(S_T - u)^+ \, du$ is a continuum of calls with strikes $u \ge K_0$, again weighted by curvature. These provide upside convexity.

In summary: the constant fixes the level, the linear term supplies the tilt, and the curvature terms supply convexity through out-of-the-money options. Any sufficiently smooth terminal payoff can therefore be expressed as a static combination of a bond, a forward, and a strip of vanilla calls and puts.

Many exotics (barriers, lookbacks with triggers, etc.) specify values on a space-time boundary $\mathcal{B}$ that surrounds $(t = 0, S_0)$ (e.g., the expiration line plus knock-in/out surfaces). The *static boundary matching principle* is

> If a portfolio Π has the same value as the target option X *at all points on* $\mathcal{B}$, and Π introduces no additional cashflows inside $\mathcal{B}$, then $V^{\Pi}(t,S) = V^{X}(t,S)$ for all (t,S) inside $\mathcal{B}$.

Operationally: choose a grid of times $0 < t_1 < \cdots < t_m < T$ and boundary levels (upper $U(\cdot)$, lower $L(\cdot)$). Build a strip of vanillas expiring at T to match the terminal payoff, then work backward in time: at t_m, add calls (for upper boundaries) with strikes $\ge U(t_m)$ and puts (for lower boundaries) with strikes $\le L(t_m)$ so that the *portfolio's theoretical value on the boundary equals the target's boundary value*. Repeat for $t_{m-1}, \ldots, t_1$. The weights are *fixed* from inception.

3 Up-and-Out Call: Example

Consider an up-and-out European call with strike K, barrier $B > K$, maturity T, zero rebate.

Step 1 (expiration line). Start with a long vanilla call $C(K,T)$; this matches the knockout's payoff *if the barrier is never hit.*

Step 2 (upper barrier at selected times). Choose times $0 < t_1 < \cdots < t_m < T$. Let $V^{(0)}(t_i,B)$ be the theoretical value, under your pricing model, of the current portfolio *on the barrier* at time t_i when $S = B$. For a true up-and-out call the boundary value is 0, so we add positions in calls struck on/above the barrier (to avoid interfering with interior cashflows). A convenient choice is to use calls struck at B expiring at t_i:

$$\Pi \leftarrow \Pi + w_i\,C(B,t_i), \qquad \text{with } w_i \text{ chosen s.t. } V^{(0)}(t_i,B) + w_i\,C(B,t_i;S=B) = 0. \tag{2}$$

If $C(B,t_i;S = B) > 0$, the weight is $w_i = -V^{(0)}(t_i,B)/C(B,t_i;S = B)$. Any strike $\geq B$ works; using B concentrates the effect *right on the boundary.*

Step 3 (iterate). Recompute $V^{(1)}$ including the new terms; proceed to t_{i-1} and solve the analogous equation. After processing all $\{t_i\}$ the portfolio has zero value at those barrier–time points and matches the vanilla call on the expiration line. By the boundary principle, the portfolio closely tracks the up-and-out call across (t,S); increasing $\{t_i\}$ density improves accuracy.

Notes.

- Use *calls* (strikes $\geq B$) for an upper boundary, *puts* (strikes $\leq L$) for a lower boundary.
- On hitting the barrier, *unwind* the static portfolio and replace it with the boundary payoff (here, zero or the specified rebate).
- The biggest dollar mismatch tends to occur near the barrier close to expiry; more boundary points in that region improve the fit.

Parity (any barrier direction)

For European barriers with zero rebate at maturity T and the same K,B:

$$\text{vanilla option} = \text{knock-in} + \text{knock-out}. \tag{3}$$

Thus a static hedge for a knock-out yields a price for the knock-in, and vice versa.

4 General Strategy

1. **Classify the payoff.**
   - *Terminal-only* $g(S_T)$: match values on the expiration line $t = T$.
   - *Path-dependent with boundaries* (e.g. barriers): match values on $t = T$ and on the barrier surfaces at selected times.
2. **Terminal match (all cases).** Use the static identity (given in (1)) to express $g(S_T)$ as cash + forward + strips of OTM vanillas at T; discretise strikes $\{K_i\}$.
3. **Boundary match (if applicable).** Choose times $0 < t_1 < \cdots < t_m < T$ and boundary levels $U(\cdot)$,$L(\cdot)$. At each t_i:

   3.1. Compute the current portfolio value on the boundary (e.g. at $S = U(t_i)$ or $S = L(t_i)$).

   3.2. Add vanillas expiring at t_i with strikes outside the interior (calls for upper boundaries, puts for lower) and solve weights so the portfolio equals the target's boundary value at those points (optionally by least squares across several nodes).
4. **Fix weights at inception.** The portfolio is static until maturity or barrier hit. On a barrier event, unwind the hedge and replace it with the boundary payoff/rebate.
5. **Implementation notes.** Use a smile-consistent desk model to value both the boundary and the hedge; allocate more boundary times and strikes near barriers and near T to reduce errors.